

A model of homophilic growth in higher-order networks

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We can think of the United States Congress as a HON.

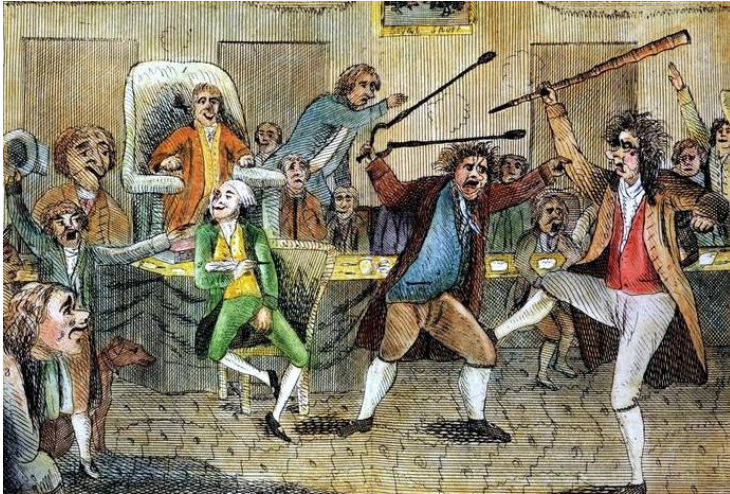
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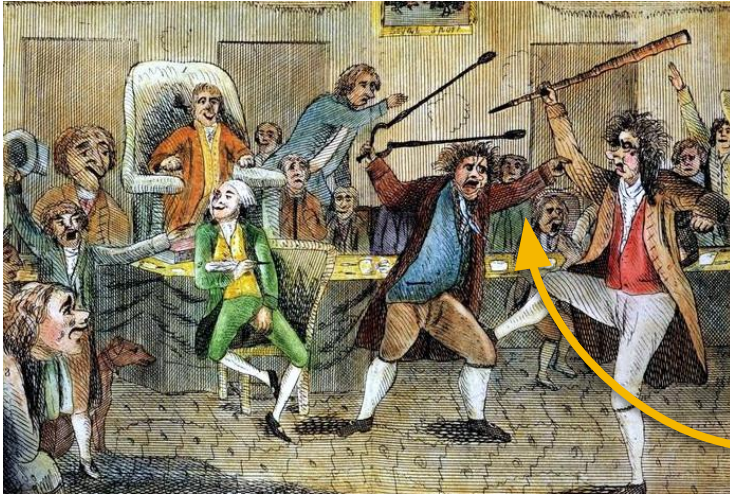
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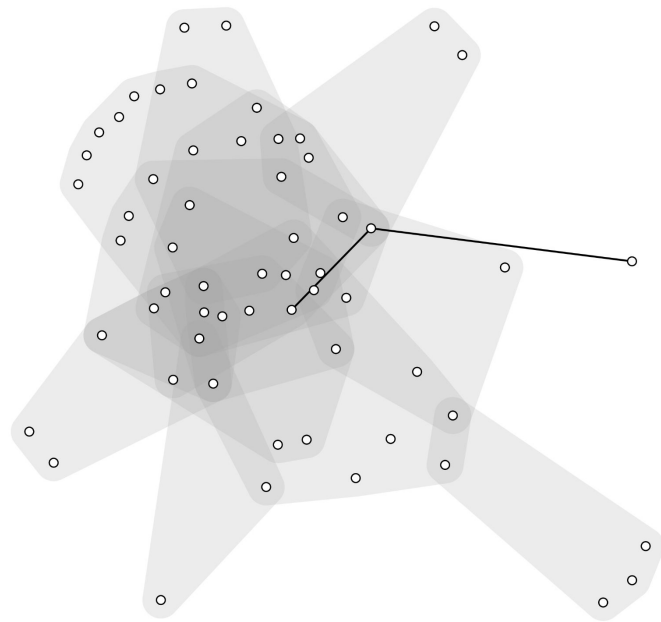


higher-order interaction!

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Hypergraphs are useful data structures for representing HONs.

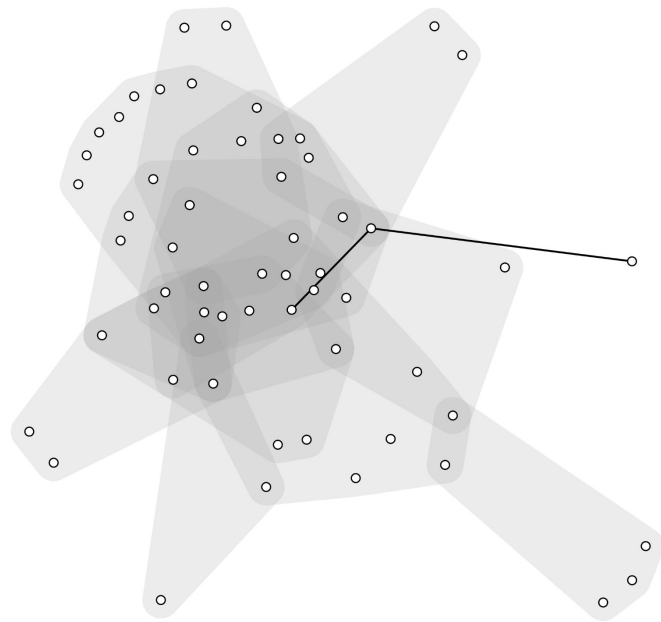


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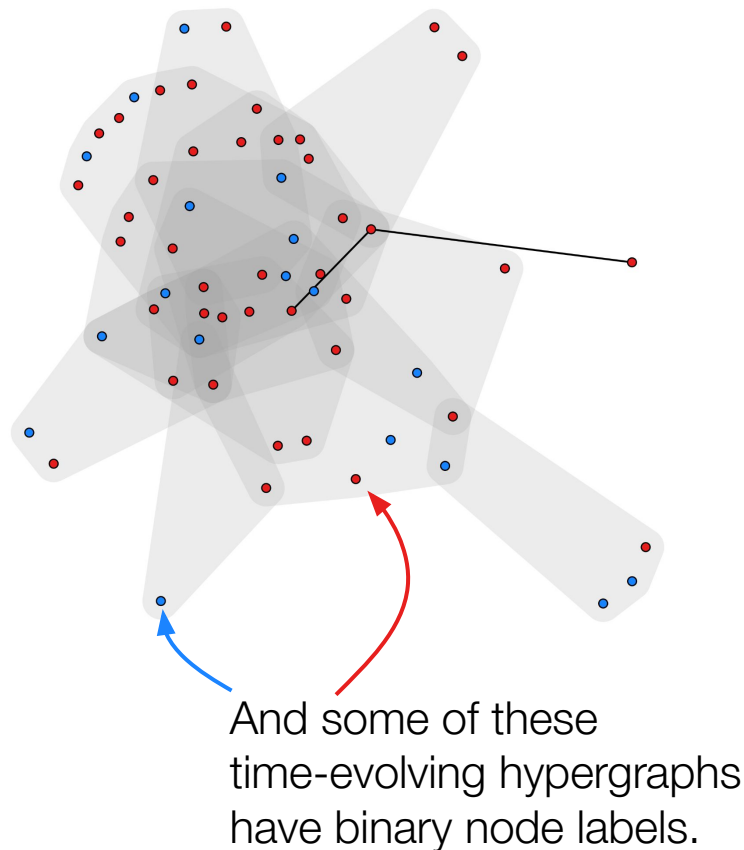


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We can think of the United States Congress as a HON.

Hypergraphs are useful data structures for representing HONs.

Some hypergraphs evolve over time.

We wonder how homophily with respect to the node labels influences the evolution process.

homophily: like
attracts like

We propose a model of
homophilic hypergraph growth.

We propose a model of homophilic hypergraph growth

This allows us to...

- Investigate the dynamics of homophilic growth in higher order networks.
- Infer parameters that describe homophily for real-world networks.

Team



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Middlebury College



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Team

Edge correlations and link prediction in growing hypergraphs

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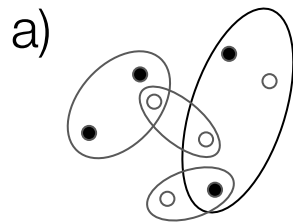


Frannie Cataldo

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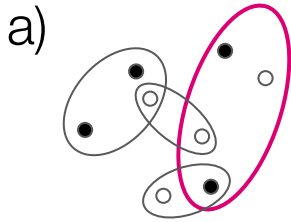
The Model

At each timestep t :

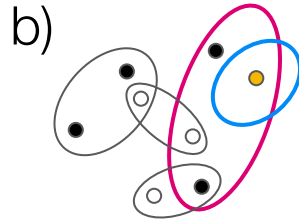
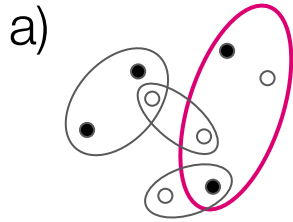


At each timestep t :

- a) Select an edge e uniformly at random.

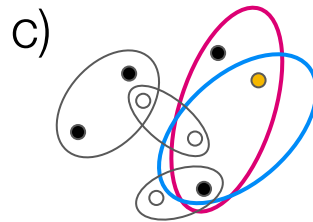
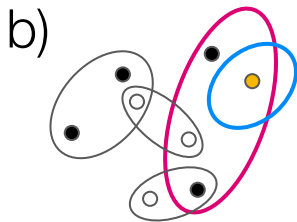
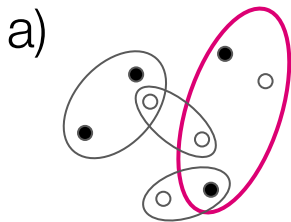


b) Select a “pivot node” u in e uniformly at random. Create a new edge e' and add u to it.



The edge copy step

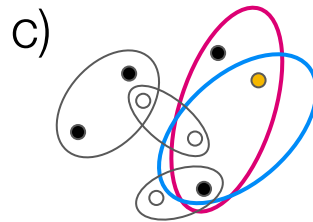
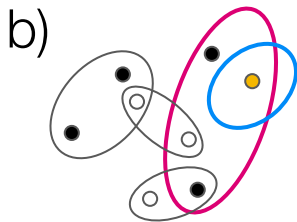
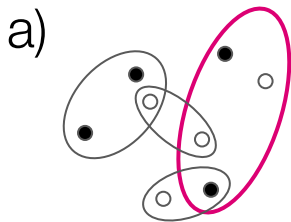
c) Add the other nodes from e to e' with probability p if they share the same node label as u and probability q otherwise.



The edge copy step

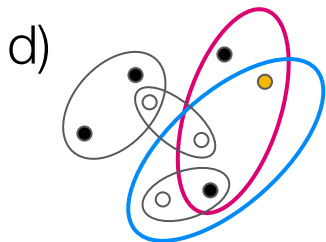
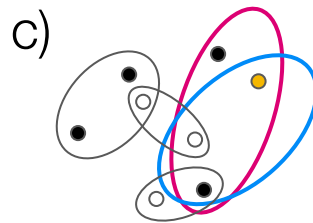
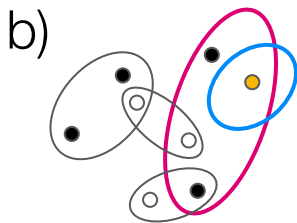
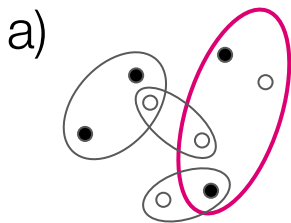
c) Add the other nodes from e to e' with probability p if they share the same node label as u and probability q otherwise.

Homophily arises when $p > q$



The external node addition step

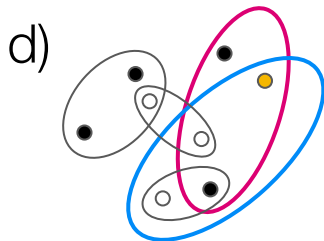
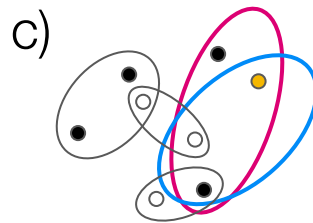
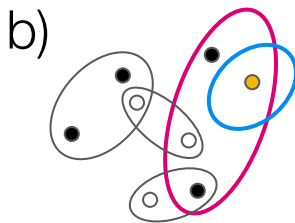
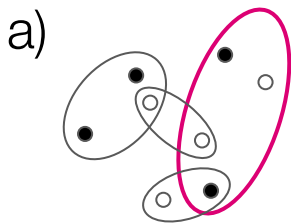
d) Generate a quantity from a Poisson distribution with parameter γ_{e,z_u} and add that many nodes outside of **e** with the same label as **u** to **e'**. Generate a value from a Poisson with parameter $\gamma_{e,\bar{z}_u}$ and add that many outside of **e** with the opposite label.



The external node addition step

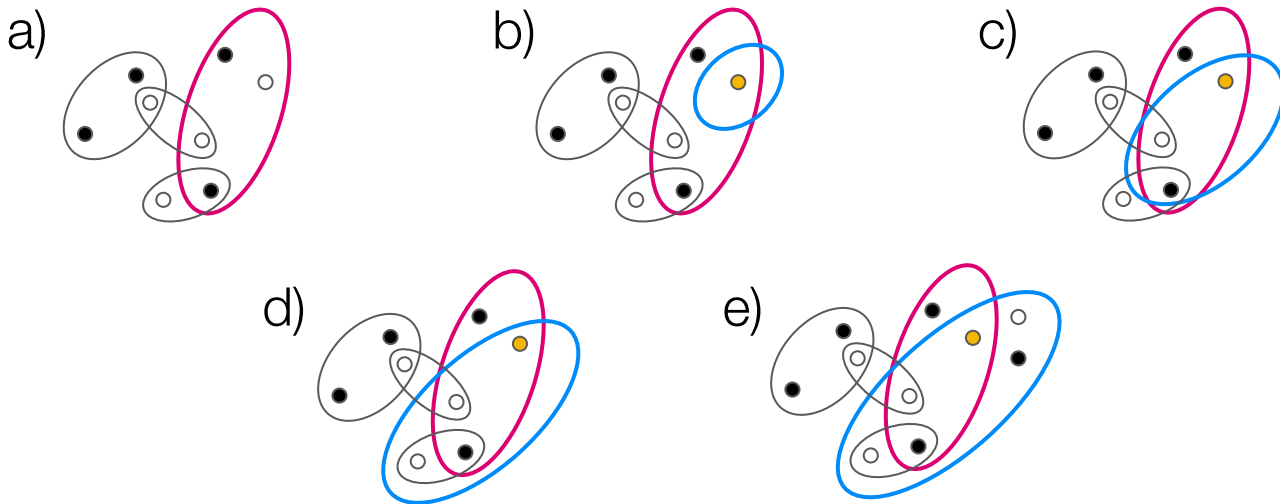
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Homophily arises when $\gamma_{e,z_u} > \gamma_{e,\bar{z}_u}$



The novel node addition step

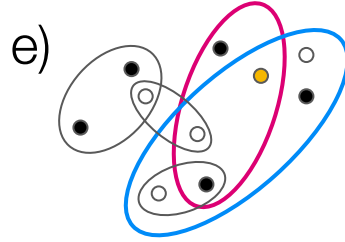
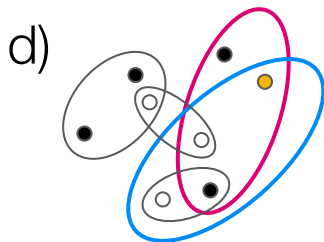
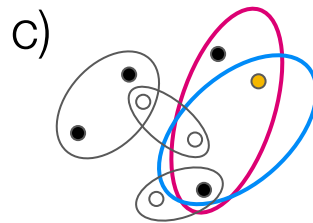
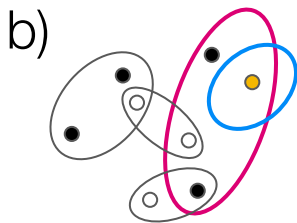
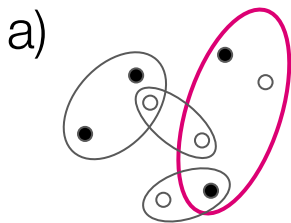
e) Generate a quantity from a Poisson distribution with parameter γ_{n,z_u} and add that many new nodes to the hypergraph, assign them the same label as u , and add them to e' . Do the same process to add new nodes of the opposite label, using a Poisson with parameter $\gamma_{n,\bar{z}_u}$.



The novel node addition step

e) Generate a quantity from a Poisson distribution with parameter γ_{n,z_u} and add that many new nodes to the hypergraph, assign them the same label as u , and add them to e' . Do the same process to add new nodes of the opposite label, using a Poisson with parameter $\gamma_{n,\bar{z}_u}$.

Homophily arises when $\gamma_{n,z_u} > \gamma_{n,\bar{z}_u}$



Dynamics

Dynamics



AKA: what can we say about
a hypergraph grown using this
model?

Power Law Degree Distribution

Assume that the degree of each node is independent of the ratio of 0-labeled nodes in the edge that was sampled to form it.

The degree distribution of a hypergraph generated by our model follows a power law.

We can derive its exponent in terms of our model parameters.

$$\zeta = 1 + \frac{\langle k \rangle}{1 + p(\rho_{00} + \rho_{11} - 1) + 2q\rho_{01}}$$

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle$$

$\langle \rangle$: mean

k : number of nodes in edge

k_i : number of nodes of label i in edge

Power Law Degree Distribution

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$$\zeta = 1 + \frac{\langle k \rangle}{1 + \underbrace{p(\rho_{00} + \rho_{11} - 1)}_{\text{copy parameters}} + \underbrace{2q\rho_{01}}_{\text{copy parameters}}}$$

$$\rho_{ij} = \left\langle \frac{k_i k_j}{k} \right\rangle$$

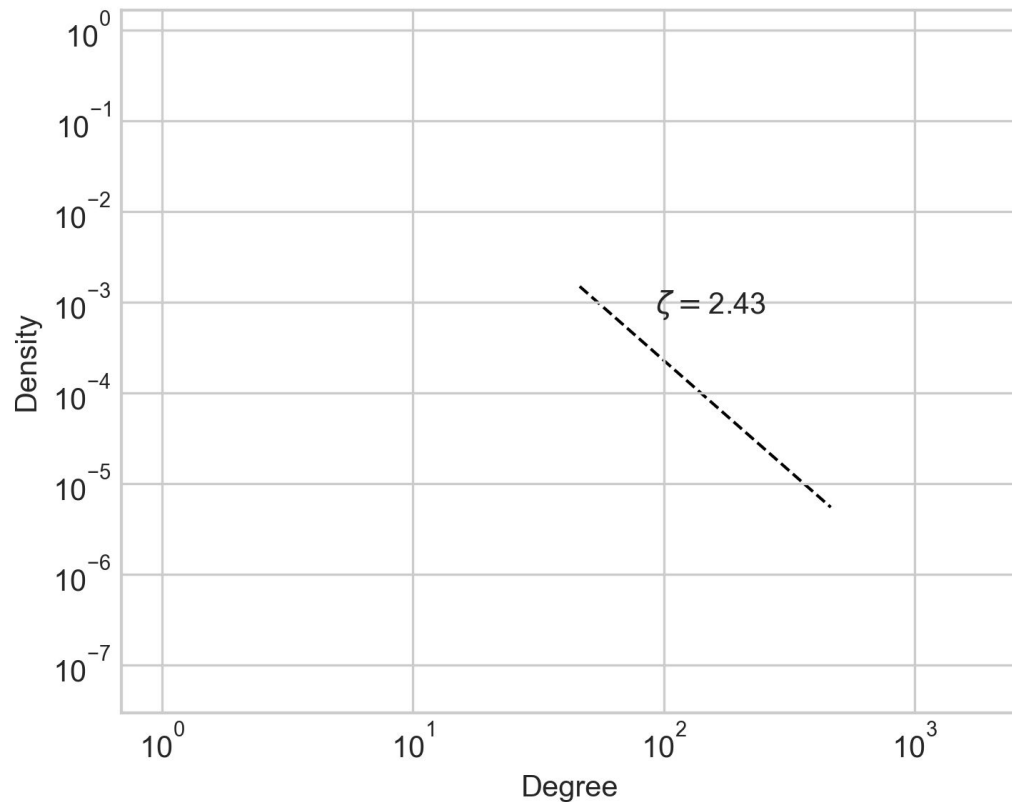
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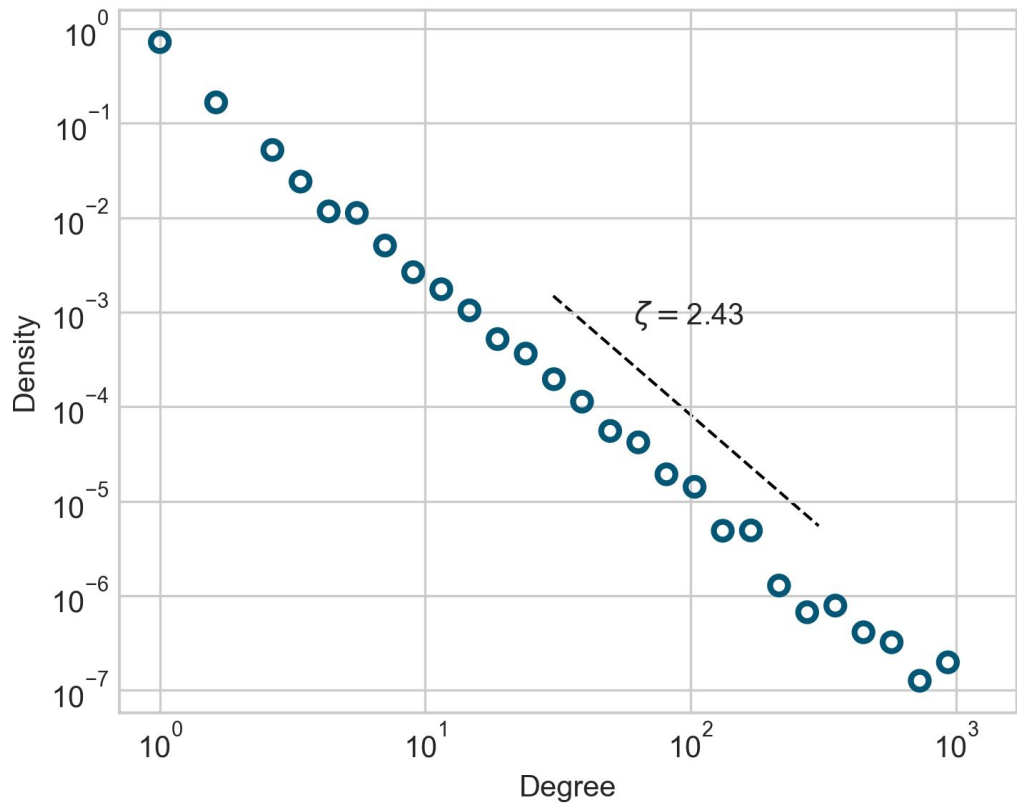
copy parameters

Power Law Degree Distribution



$$\begin{aligned} p &= 0.7, q = 0.3, \\ \gamma_{e,z_u} &= 0.5, \gamma_{e,\bar{z}_u} = 0.3, \\ \gamma_{n,z_u} &= 0.2, \gamma_{n,\bar{z}_u} = 0.1 \end{aligned}$$

Power Law Degree Distribution



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Inference

Given a hypergraph, can we determine what parameters generated it?

How **homophilic** is the formation of new connections?

How strong is the edge copy **mechanism**?

Maximum Likelihood Estimation

Maximizes over the log marginal likelihood by **computing the likelihood itself**

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Expectation Maximization

Maximizes over the log marginal likelihood by **computing expected sufficient statistics**

Avoids marginalizing over latent variables



Maximum Likelihood Estimation

Maximizes over the log marginal likelihood by computing the likelihood itself



Expectation Maximization

Maximizes over the log marginal likelihood by computing expected sufficient statistics

Avoids marginalizing over latent variables



Stochastic Expectation Maximization

Maximizes over the log marginal likelihood by computing expected sufficient statistics *given a single observation*

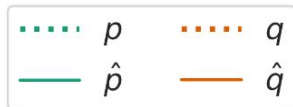
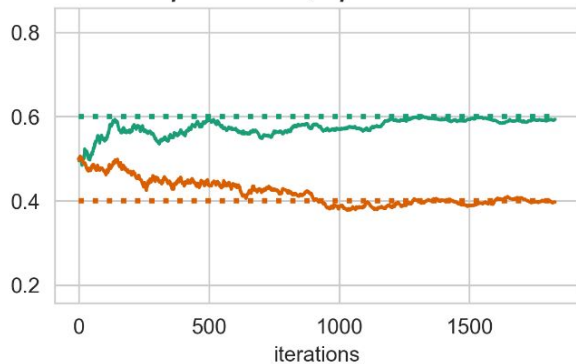
Avoids marginalizing over latent variables and computing expectations over the complete data



SEM performs well on synthetic data

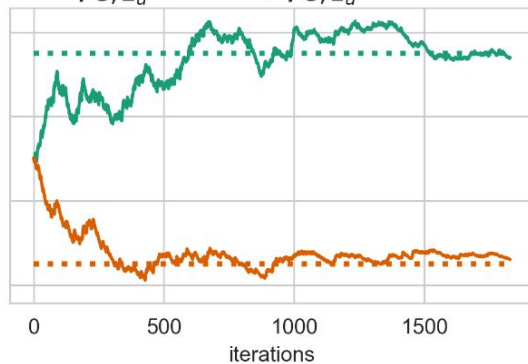
Edge copy step

$$\hat{p} = 0.59, \hat{q} = 0.40$$



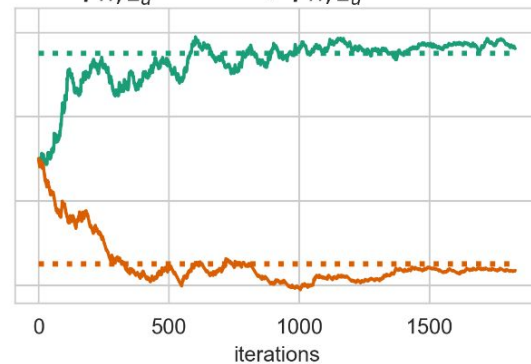
External node step

$$\hat{\gamma}_{e, z_u} = 0.74, \hat{\gamma}_{e, \bar{z}_u} = 0.26$$



Novel node step

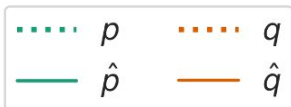
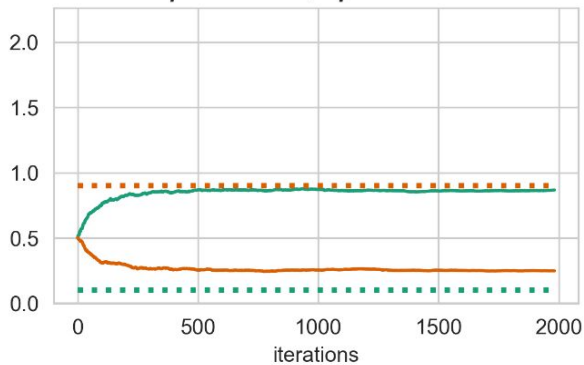
$$\hat{\gamma}_{n, z_u} = 0.76, \hat{\gamma}_{n, \bar{z}_u} = 0.23$$



... with one exception

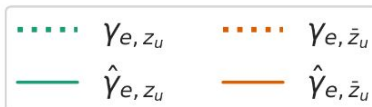
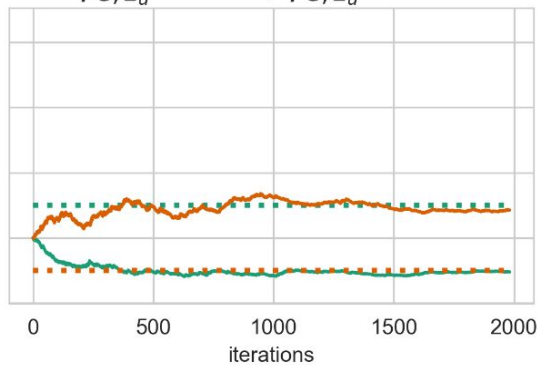
Edge copy step

$$\hat{p} = 0.87, \hat{q} = 0.25$$



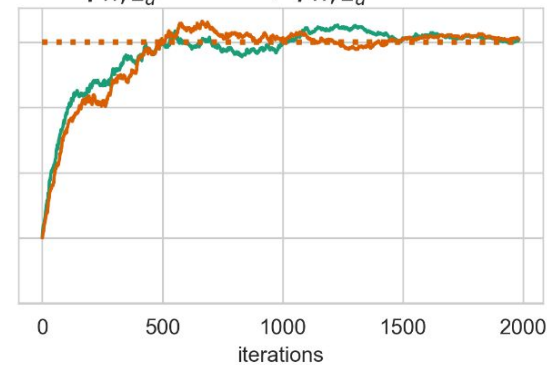
External node step

$$\hat{\gamma}_{e, z_u} = 0.24, \hat{\gamma}_{e, \bar{z}_u} = 0.71$$



Novel node step

$$\hat{\gamma}_{n, z_u} = 2.02, \hat{\gamma}_{n, \bar{z}_u} = 2.02$$



It struggles when the out-group parameter is higher than the in-group one

Inference using our model tells us about the properties of real-world higher-order networks.

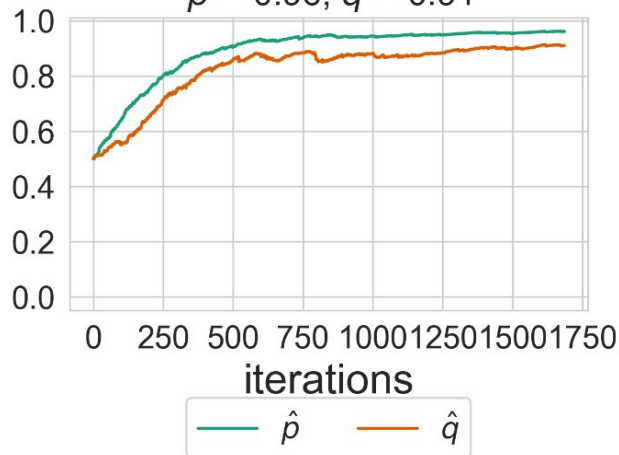
Some hypergraphs exhibit little to no homophily when forming new connections

data: social interactions between high school students

binary labeling: gender M/F

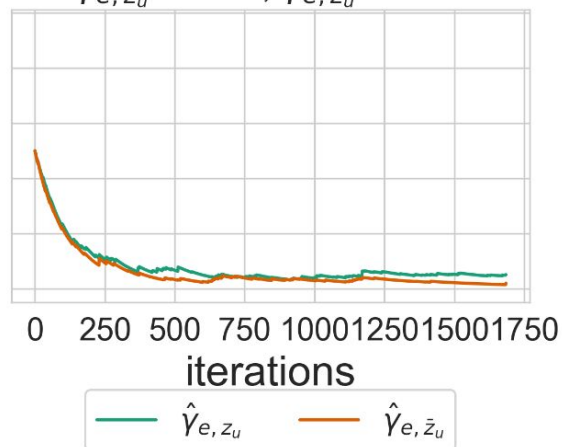
Edge copy step

$$\hat{p} = 0.96, \hat{q} = 0.91$$



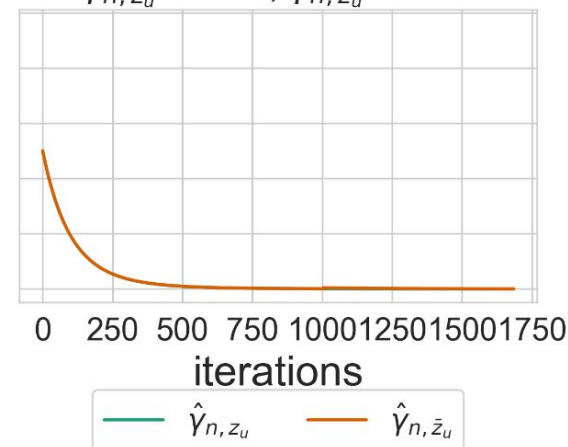
External node step

$$\hat{\gamma}_{e, z_u} = 0.05, \hat{\gamma}_{e, \bar{z}_u} = 0.02$$



Novel node step

$$\hat{\gamma}_{n, z_u} = 0.00, \hat{\gamma}_{n, \bar{z}_u} = 0.00$$



Some hypergraphs exhibit little to no homophily when forming new connections

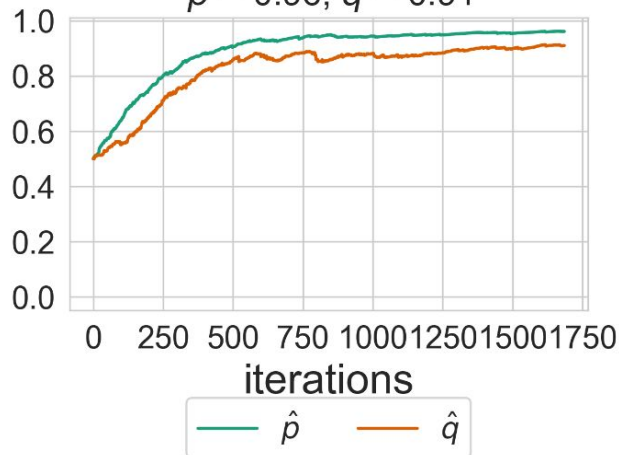
data: social interactions between high school students

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Note #1: This result is dependent on the choice of label

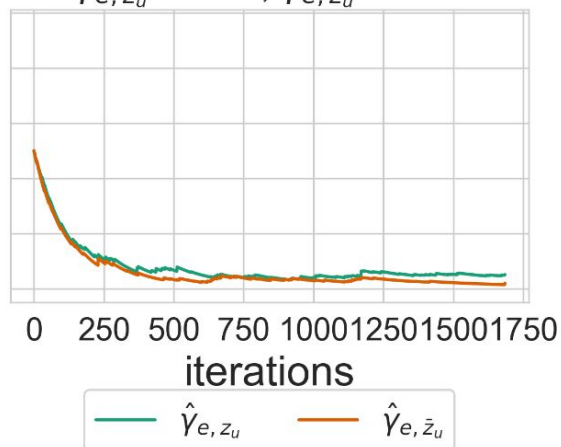
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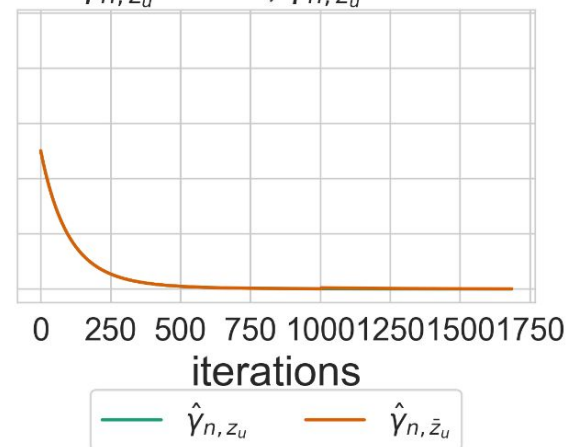
External node step

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Novel node step

$$\hat{\gamma}_{n, z_u} = 0.00, \hat{\gamma}_{n, \bar{z}_u} = 0.00$$



Some hypergraphs exhibit little to no homophily when forming new connections

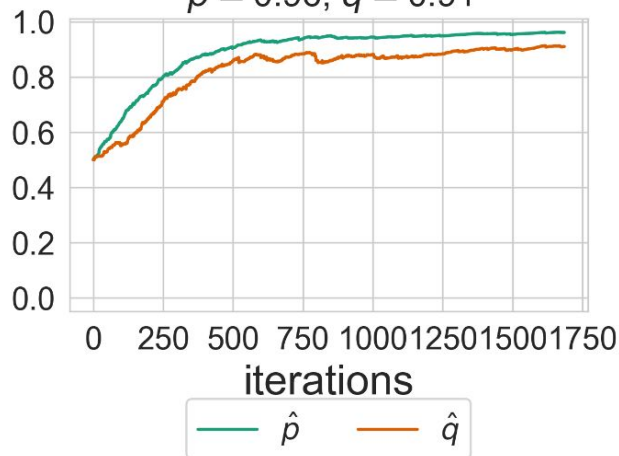
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Note #2: Lack of homophily $\neq$ equal representation (mean proportion of the within-edge majority is 0.85)

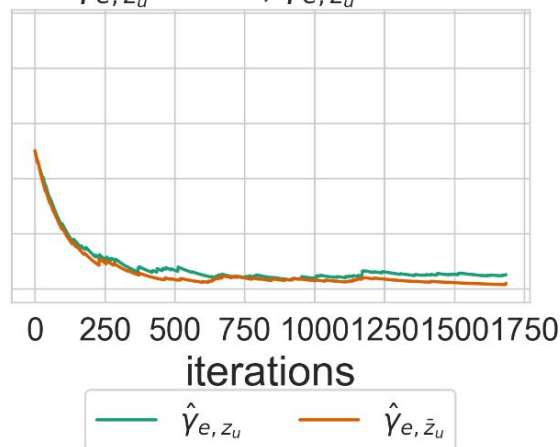
Edge copy step

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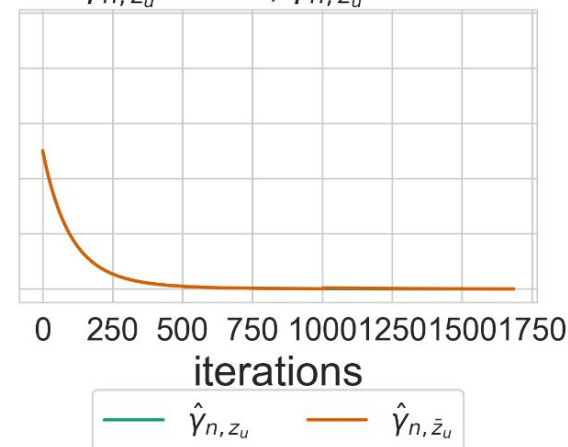
External node step

$$\hat{\gamma}_{e, z_u} = 0.05, \hat{\gamma}_{e, \bar{z}_u} = 0.02$$



Novel node step

$$\hat{\gamma}_{n, z_u} = 0.00, \hat{\gamma}_{n, \bar{z}_u} = 0.00$$



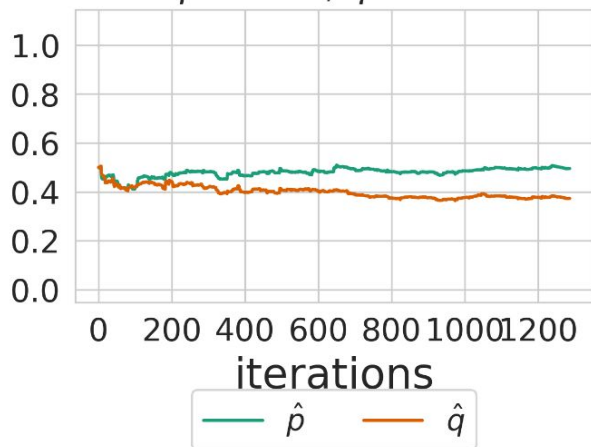
Homophily is evident in the growth processes of others

data: bill cosponsorship in the United States Senate

binary labeling: political party D/R

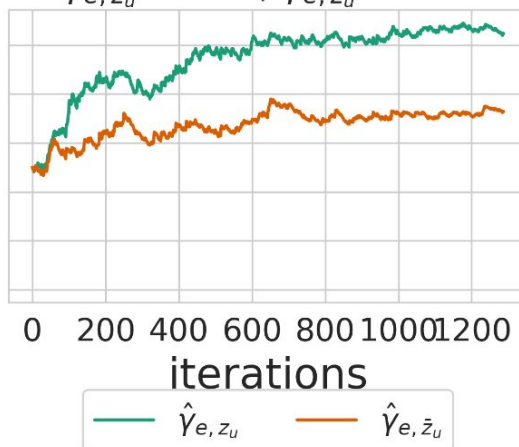
Edge copy step

$$\hat{p} = 0.50, \hat{q} = 0.37$$



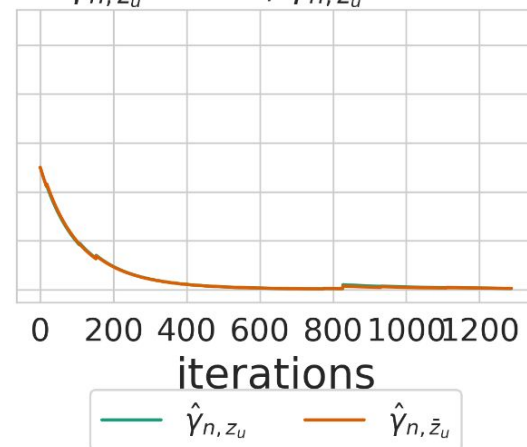
External node step

$$\hat{\gamma}_{e,z_u} = 1.05, \hat{\gamma}_{e,\bar{z}_u} = 0.73$$



Novel node step

$$\hat{\gamma}_{n,z_u} = 0.00, \hat{\gamma}_{n,\bar{z}_u} = 0.00$$



Conclusions

We proposed a model of hypergraph growth where homophily can appear in three different ways:

1. Edge copying
2. External node addition
3. Novel node addition

We can describe the dynamics of hypergraphs that grow in this way:

- They have power law distributed node degrees with known exponent

We can infer the model parameters using stochastic expectation maximization.

- This provides insights into the evolution dynamics of real-world higher order networks.

Discussion

There are many ways to investigate the properties of a higher-order network with respect to its node labels.

Often, these are **structural** (e.g. community detection)

Structural metrics tell us about the final state of a hypergraph

Our model informs on the **mechanisms that produce that final hypergraph**

So, in addition to saying “this network is segregated”

we can also say “**this is how the segregation emerged**”

Thanks + Acknowledgements



Phil Chodrow
Middlebury College



Frannie Cataldo
Middlebury College



Dan Larremore
University of Colorado Boulder



My website



Supplementary Slides

Maximum Likelihood Estimation

In MLE, our parameter estimates are the values θ that maximize the log marginal likelihood

$$\mathcal{L}(\theta) = \sum_{i=1}^N \log p(x_i|\theta)$$

The problem: **latent variables** (u and e)

Expectation Maximization

In EM, we harness the power of the sufficient statistic.

Informally: T is a sufficient statistic if it tells a scientist all of the same information about the θ as the complete data do.

At each iteration:

E step: Compute the expected sufficient statistics given the complete data.
Update the sufficient statistic estimates.

M step: Use the updated sufficient statistics to compute the MLEs of the parameters.

The problem: **computing expectations over the complete data is expensive**

Stochastic Expectation Maximization

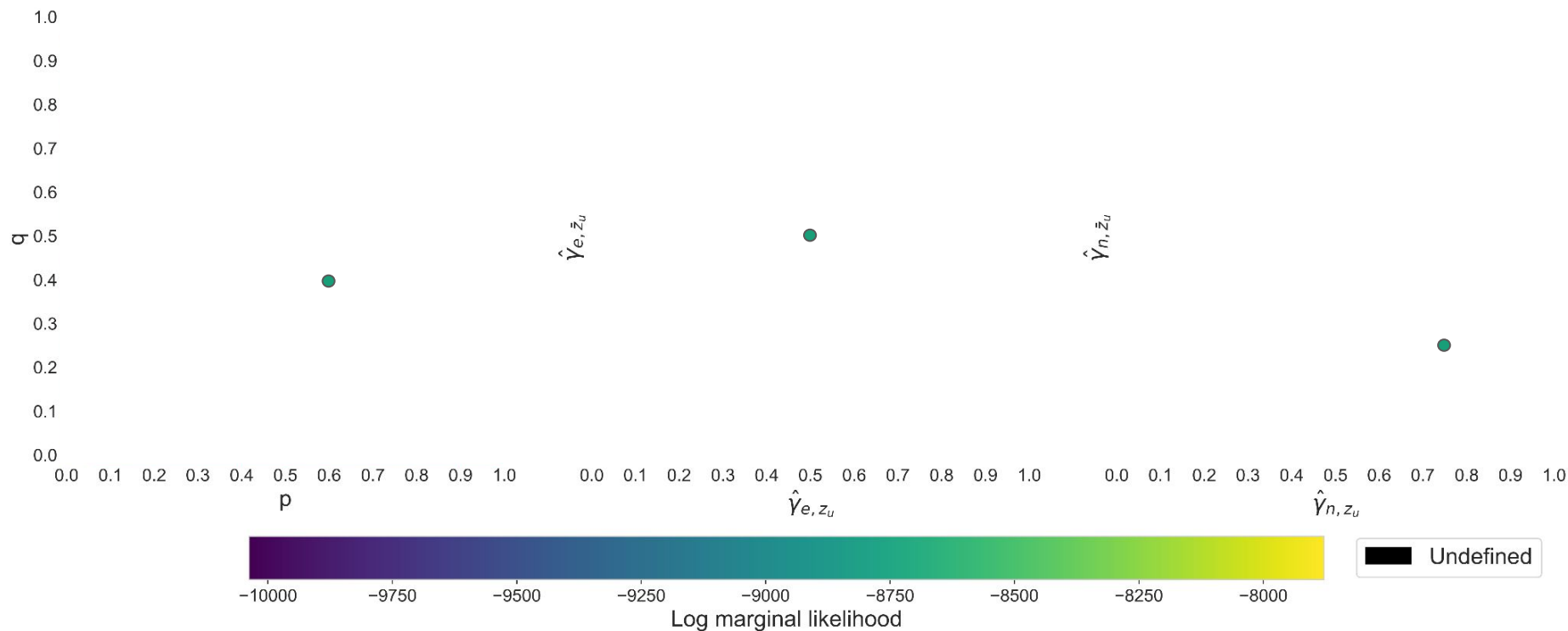
In SEM, we use only one observation x_i at each iteration.

At each iteration:

E step: Compute the expected sufficient statistics given x_i . Update the sufficient statistic estimates.

M step: Use the updated sufficient statistics to compute the MLEs of the parameters.

A quick sanity check



A quick sanity check

